SCR2043 OPERATING SYSTEMS

Name	:	AIN NURNABILA BINTI MOHD AZHAR	Marks
Student ID	:	A23CS0207	
Section	:	2	

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to process monitoring and management in operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment02-studentname-matricno.docx**.

Essential Steps Before Starting Lab Assessment 2:

1. **Download necessary source codes:**

   Use the `wget` command to retrieve the following source code files to your Linux (or WSL or MacOS) environment:

```
wget -O mainprocess.c https://rebrand.ly/mainprocess_c
wget -O subprocess1.c https://rebrand.ly/subprocess1_c
wget -O subprocess2.c https://rebrand.ly/subprocess2_c
```

2. **Compile the source files:**

   Use the `gcc` compiler to create executable files from the source code.

```
gcc mainprocess.c -o mainprocess
gcc subprocess1.c -o subprocess1
gcc subprocess2.c -o subprocess2
```

3. **Execute the dummy processes:**

   Run all the dummy processes

```
./mainprocess &
```

   Press **enter** two times.

4. The dummy processes are running for 2 hours. If you took longer than 2 hours on questions 1-9, please restart the main process with `./mainprocess &`.

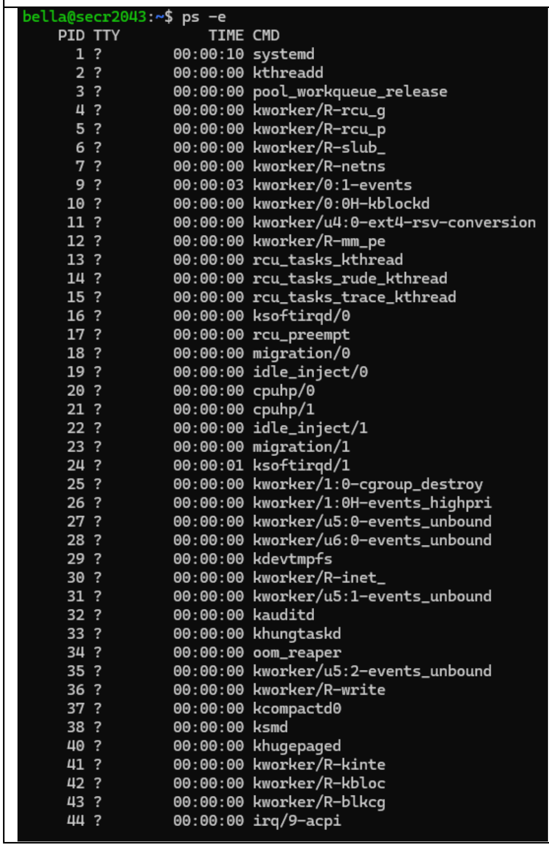
Lab Assessment 2 : Linux Process Monitoring and Management

Instructions:

1. Carefully execute each command as instructed in the questions.
2. Write down the exact command used for each task.
3. Capture a screenshot of the command's output.

Question 1

Use the `ps` command with the appropriate option to display a complete list of all running processes within the Linux operating system.

Command
`ps -e`
Output


```
bella@secr2043:~$ ps -e
    PID TTY          TIME CMD
      1 ?        00:00:10 systemd
      2 ?        00:00:00 kthreadd
      3 ?        00:00:00 pool_workqueue_release
      4 ?        00:00:00 kworker/R-rcu_g
      5 ?        00:00:00 kworker/R-rcu_p
      6 ?        00:00:00 kworker/R-slub_
      7 ?        00:00:00 kworker/R-netns
      9 ?        00:00:03 kworker/0:1-events
     10 ?        00:00:00 kworker/0:0H-kblockd
     11 ?        00:00:00 kworker/u4:0-ext4-rsv-conversion
     12 ?        00:00:00 kworker/R-mm_pe
     13 ?        00:00:00 rcu_tasks_kthread
     14 ?        00:00:00 rcu_tasks_rude_kthread
     15 ?        00:00:00 rcu_tasks_trace_kthread
     16 ?        00:00:00 ksoftirqd/0
     17 ?        00:00:00 rcu_preempt
     18 ?        00:00:00 migration/0
     19 ?        00:00:00 idle_inject/0
     20 ?        00:00:00 cpuhp/0
     21 ?        00:00:00 cpuhp/1
     22 ?        00:00:00 idle_inject/1
     23 ?        00:00:00 migration/1
     24 ?        00:00:01 ksoftirqd/1
     25 ?        00:00:00 kworker/1:0-cgroup_destroy
     26 ?        00:00:00 kworker/1:0H-events_highpri
     27 ?        00:00:00 kworker/u5:0-events_unbound
     28 ?        00:00:00 kworker/u6:0-events_unbound
     29 ?        00:00:00 kdevtmpfs
     30 ?        00:00:00 kworker/R-inet_
     31 ?        00:00:00 kworker/u5:1-events_unbound
     32 ?        00:00:00 kauditd
     33 ?        00:00:00 khungtaskd
     34 ?        00:00:00 oom_reaper
     35 ?        00:00:00 kworker/u5:2-events_unbound
     36 ?        00:00:00 kworker/R-write
     37 ?        00:00:00 kcompactd0
     38 ?        00:00:00 ksmd
     40 ?        00:00:00 khugepaged
     41 ?        00:00:00 kworker/R-kinte
     42 ?        00:00:00 kworker/R-kbloc
     43 ?        00:00:00 kworker/R-blkcg
     44 ?        00:00:00 irq/9-acpi
```

```
   45 ?        00:00:00 kworker/R-tpm_d
   46 ?        00:00:00 kworker/R-ata_s
   47 ?        00:00:00 kworker/R-md
   48 ?        00:00:00 kworker/R-md_bi
   49 ?        00:00:00 kworker/R-edac-
   50 ?        00:00:00 kworker/R-devfr
   51 ?        00:00:00 watchdogd
   52 ?        00:00:00 kworker/u6:1-events_power_efficient
   55 ?        00:00:00 kswapd0
   56 ?        00:00:00 ecryptfs-kthread
   57 ?        00:00:00 kworker/R-kthro
   58 ?        00:00:00 kworker/R-acpi_
   59 ?        00:00:00 scsi_eh_0
   60 ?        00:00:00 kworker/R-scsi_
   61 ?        00:00:00 scsi_eh_1
   62 ?        00:00:00 kworker/R-scsi_
   64 ?        00:00:00 kworker/u5:3
   65 ?        00:00:00 kworker/R-mld
   66 ?        00:00:00 kworker/R-ipv6_
   73 ?        00:00:00 kworker/R-kstrp
   75 ?        00:00:00 kworker/u7:0
   76 ?        00:00:00 kworker/u8:0
   77 ?        00:00:00 kworker/u9:0
   82 ?        00:00:00 kworker/R-crypt
   91 ?        00:00:00 kworker/R-charg
  130 ?        00:00:00 kworker/u4:1
  132 ?        00:00:00 kworker/0:3-cgroup_destroy
  133 ?        00:00:00 kworker/R-mpt_p
  134 ?        00:00:00 kworker/R-mpt/0
  154 ?        00:00:00 scsi_eh_2
  155 ?        00:00:00 kworker/R-scsi_
  156 ?        00:00:00 kworker/u6:2-events_power_efficient
  158 ?        00:00:05 kworker/1:2-events
  191 ?        00:00:00 kworker/R-raid5
  228 ?        00:00:00 kworker/1:2H-kblockd
  229 ?        00:00:00 jbd2/sda1-8
  230 ?        00:00:00 kworker/R-ext4-
  297 ?        00:00:00 psimon
  302 ?        00:00:01 systemd-journal
  339 ?        00:00:00 kworker/R-kmpat
  340 ?        00:00:00 kworker/R-kmpat
  362 ?        00:00:01 multipathd
  365 ?        00:00:01 systemd-udevd
  369 ?        00:00:04 kworker/0:4-events
  371 ?        00:00:00 psimon
  421 ?        00:00:00 kworker/0:2H-kblockd
  439 ?        00:00:00 irq/18-vmwgfx
  440 ?        00:00:00 kworker/R-ttm
  484 ?        00:00:00 jbd2/sda16-8
  485 ?        00:00:00 kworker/R-ext4-
  512 ?        00:00:01 systemd-network
  525 ?        00:00:02 kworker/1:3-mm_percpu_wq
  540 ?        00:00:00 systemd-resolve
  543 ?        00:00:00 systemd-timesyn
  645 ?        00:00:00 dbus-daemon
  649 ?        00:00:00 polkitd
  655 ?        00:00:00 systemd-logind
  657 ?        00:00:00 udisksd
  677 ?        00:00:00 rsyslogd
  689 ?        00:00:01 unattended-upgr
  719 ?        00:00:00 cron
  722 ?        00:00:00 ModemManager
  751 tty1     00:00:00 agetty
  850 ?        00:00:00 sshd
  852 ?        00:00:00 sshd
  859 ?        00:00:04 sshd
  860 pts/0    00:00:00 bash
  900 pts/0    00:00:00 ps
```

Question 2

Employ the `ps` command with necessary options to unveil comprehensive details about each running process.

Command
ps -ef
Output

```
bella@secr2043:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  2 14:36 ?        00:00:10 /sbin/init
root           2       0  0 14:36 ?        00:00:00 [kthreadd]
root           3       2  0 14:36 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 14:36 ?        00:00:00 [kworker/R-rcu_g]
root           5       2  0 14:36 ?        00:00:00 [kworker/R-rcu_p]
root           6       2  0 14:36 ?        00:00:00 [kworker/R-slub_]
root           7       2  0 14:36 ?        00:00:00 [kworker/R-netns]
root           9       2  0 14:36 ?        00:00:03 [kworker/0:1-events]
root          10       2  0 14:36 ?        00:00:00 [kworker/0:0H-kblockd]
root          11       2  0 14:36 ?        00:00:00 [kworker/u4:0-ext4-rsv-conversion]
root          12       2  0 14:36 ?        00:00:00 [kworker/R-mm_pe]
root          13       2  0 14:36 ?        00:00:00 [rcu_tasks_kthread]
root          14       2  0 14:36 ?        00:00:00 [rcu_tasks_rude_kthread]
root          15       2  0 14:36 ?        00:00:00 [rcu_tasks_trace_kthread]
root          16       2  0 14:36 ?        00:00:00 [ksoftirqd/0]
root          17       2  0 14:36 ?        00:00:00 [rcu_preempt]
root          18       2  0 14:36 ?        00:00:00 [migration/0]
root          19       2  0 14:36 ?        00:00:00 [idle_inject/0]
root          20       2  0 14:36 ?        00:00:00 [cpuhp/0]
root          21       2  0 14:36 ?        00:00:00 [cpuhp/1]
root          22       2  0 14:36 ?        00:00:00 [idle_inject/1]
root          23       2  0 14:36 ?        00:00:00 [migration/1]
root          24       2  0 14:36 ?        00:00:01 [ksoftirqd/1]
root          25       2  0 14:36 ?        00:00:00 [kworker/1:0-cgroup_destroy]
root          26       2  0 14:36 ?        00:00:00 [kworker/1:0H-events_highpri]
root          27       2  0 14:36 ?        00:00:00 [kworker/u5:0-events_power_efficient]
root          28       2  0 14:36 ?        00:00:00 [kworker/u6:0-events_unbound]
root          29       2  0 14:36 ?        00:00:00 [kdevtmpfs]
root          30       2  0 14:36 ?        00:00:00 [kworker/R-inet_]
root          31       2  0 14:36 ?        00:00:00 [kworker/u5:1-events_unbound]
root          32       2  0 14:36 ?        00:00:00 [kauditd]
root          33       2  0 14:36 ?        00:00:00 [khungtaskd]
root          34       2  0 14:36 ?        00:00:00 [oom_reaper]
root          35       2  0 14:36 ?        00:00:00 [kworker/u5:2-events_unbound]
root          36       2  0 14:36 ?        00:00:00 [kworker/R-write]
root          37       2  0 14:36 ?        00:00:00 [kcompactd0]
root          38       2  0 14:36 ?        00:00:00 [ksmd]
root          40       2  0 14:36 ?        00:00:00 [khugepaged]
root          41       2  0 14:36 ?        00:00:00 [kworker/R-kinte]
root          42       2  0 14:36 ?        00:00:00 [kworker/R-kbloc]
root          43       2  0 14:36 ?        00:00:00 [kworker/R-blkcg]
root          44       2  0 14:36 ?        00:00:00 [irq/9-acpi]
root          45       2  0 14:36 ?        00:00:00 [kworker/R-tpm_d]
root          46       2  0 14:36 ?        00:00:00 [kworker/R-ata_s]
root          47       2  0 14:36 ?        00:00:00 [kworker/R-md]
root          48       2  0 14:36 ?        00:00:00 [kworker/R-md_bi]
root          49       2  0 14:36 ?        00:00:00 [kworker/R-edac-]
root          48       2  0 14:36 ?        00:00:00 [kworker/R-md_bi]
root          49       2  0 14:36 ?        00:00:00 [kworker/R-edac-]
root          50       2  0 14:36 ?        00:00:00 [kworker/R-devfr]
root          51       2  0 14:36 ?        00:00:00 [watchdogd]
root          52       2  0 14:36 ?        00:00:00 [kworker/u6:1-events_power_efficient]
root          55       2  0 14:36 ?        00:00:00 [kswapd0]
root          56       2  0 14:36 ?        00:00:00 [ecryptfs-kthread]
root          57       2  0 14:36 ?        00:00:00 [kworker/R-kthro]
root          58       2  0 14:36 ?        00:00:00 [kworker/R-acpi_]
root          59       2  0 14:36 ?        00:00:00 [scsi_eh_0]
root          60       2  0 14:36 ?        00:00:00 [kworker/R-scsi_]
root          61       2  0 14:36 ?        00:00:00 [scsi_eh_1]
root          62       2  0 14:36 ?        00:00:00 [kworker/R-scsi_]
root          64       2  0 14:36 ?        00:00:00 [kworker/u5:3]
root          65       2  0 14:36 ?        00:00:00 [kworker/R-mld]
root          66       2  0 14:36 ?        00:00:00 [kworker/R-ipv6_]
root          73       2  0 14:36 ?        00:00:00 [kworker/R-kstrp]
root          75       2  0 14:36 ?        00:00:00 [kworker/u7:0]
root          76       2  0 14:36 ?        00:00:00 [kworker/u8:0]
root          77       2  0 14:36 ?        00:00:00 [kworker/u9:0]
root          82       2  0 14:36 ?        00:00:00 [kworker/R-crypt]
root          91       2  0 14:36 ?        00:00:00 [kworker/R-charg]
root         130       2  0 14:36 ?        00:00:00 [kworker/u4:1]
root         133       2  0 14:36 ?        00:00:00 [kworker/R-mpt_p]
root         134       2  0 14:36 ?        00:00:00 [kworker/R-mpt/0]
root         154       2  0 14:36 ?        00:00:00 [scsi_eh_2]
root         155       2  0 14:36 ?        00:00:00 [kworker/R-scsi_]
root         156       2  0 14:36 ?        00:00:00 [kworker/u6:2-events_unbound]
root         158       2  1 14:36 ?        00:00:06 [kworker/1:2-events]
root         191       2  0 14:36 ?        00:00:00 [kworker/R-raid5]
root         228       2  0 14:36 ?        00:00:00 [kworker/1:2H-kblockd]
root         229       2  0 14:36 ?        00:00:00 [jbd2/sda1-8]
root         230       2  0 14:36 ?        00:00:00 [kworker/R-ext4-]
root         297       2  0 14:36 ?        00:00:00 [psimon]
root         302       1  0 14:36 ?        00:00:01 /usr/lib/systemd/systemd-journald
root         339       2  0 14:36 ?        00:00:00 [kworker/R-kmpat]
root         340       2  0 14:36 ?        00:00:00 [kworker/R-kmpat]
root         362       1  0 14:36 ?        00:00:01 /sbin/multipathd -d -s
root         365       1  0 14:36 ?        00:00:01 /usr/lib/systemd/systemd-udevd
root         369       2  1 14:36 ?        00:00:06 [kworker/0:4-events]
root         371       2  0 14:36 ?        00:00:00 [psimon]
root         421       2  0 14:36 ?        00:00:00 [kworker/0:2H-kblockd]
root         439       2  0 14:36 ?        00:00:00 [irq/18-vmwgfx]
root         440       2  0 14:36 ?        00:00:00 [kworker/R-ttm]
root         484       2  0 14:36 ?        00:00:00 [jbd2/sda16-8]
root         485       2  0 14:36 ?        00:00:00 [kworker/R-ext4-]
systemd+     512       1  0 14:36 ?        00:00:01 /usr/lib/systemd/systemd-networkd
root         525       2  0 14:36 ?        00:00:02 [kworker/1:3-mm_percpu_wq]
systemd+     540       1  0 14:36 ?        00:00:00 /usr/lib/systemd/systemd-resolved
```

```
systemd+     543       1  0 14:36 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
message+     645       1  0 14:37 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
polkitd      649       1  0 14:37 ?        00:00:00 /usr/lib/polkit-1/polkitd --no-debug
root         655       1  0 14:37 ?        00:00:00 /usr/lib/systemd/systemd-logind
root         657       1  0 14:37 ?        00:00:00 /usr/libexec/udisks2/udisksd
syslog       677       1  0 14:37 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root         689       1  0 14:37 ?        00:00:01 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root         719       1  0 14:37 ?        00:00:00 /usr/sbin/cron -f -P
root         722       1  0 14:37 ?        00:00:00 /usr/sbin/ModemManager
root         751       1  0 14:37 tty1     00:00:00 /sbin/agetty -o -p -- \u --noclear - linux
root         850       1  0 14:37 ?        00:00:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root         852     850  0 14:37 ?        00:00:00 sshd: bella [priv]
bella        859     852  1 14:38 ?        00:00:05 sshd: bella@pts/0
bella        860     859  0 14:38 pts/0    00:00:00 -bash
root         901       2  1 14:42 ?        00:00:01 [kworker/0:0-events]
bella        902     860 99 14:44 pts/0    00:00:00 ps -ef
```

Question 3

Use the `ps` command with some tools to only list processes named "subprocess" and show some info about them.

Command
`ps -ef	grep subprocess`
Output

```
bella        912     860 66 14:47 pts/0    00:00:00 grep --color=auto subprocess
```

Question 4

Execute the `ps` command, specifying options that reveal only the following columns:

- Process ID (`pid`)
- Owner of the process (`user`)
- CPU percentage (`pcpu`)
- Memory percentage (`pmem`)
- Command (`cmd`)

Command
`ps -eo pid, user, pcpu, pmem, cmd`
Output

```
bella@secr2043:~$ ps -eo pid,user,pcpu,pmem,cmd
    PID USER     %CPU %MEM CMD
      1 root      1.2  1.3 /sbin/init
      2 root      0.0  0.0 [kthreadd]
      3 root      0.0  0.0 [pool_workqueue_release]
      4 root      0.0  0.0 [kworker/R-rcu_g]
      5 root      0.0  0.0 [kworker/R-rcu_p]
      6 root      0.0  0.0 [kworker/R-slub_]
      7 root      0.0  0.0 [kworker/R-netns]
      9 root      0.3  0.0 [kworker/0:1-events]
     10 root      0.0  0.0 [kworker/0:0H-kblockd]
     11 root      0.0  0.0 [kworker/u4:0-ext4-rsv-conversion]
     12 root      0.0  0.0 [kworker/R-mm_pe]
     13 root      0.0  0.0 [rcu_tasks_kthread]
     14 root      0.0  0.0 [rcu_tasks_rude_kthread]
     15 root      0.0  0.0 [rcu_tasks_trace_kthread]
     16 root      0.0  0.0 [ksoftirqd/0]
     17 root      0.0  0.0 [rcu_preempt]
     18 root      0.0  0.0 [migration/0]
     19 root      0.0  0.0 [idle_inject/0]
     20 root      0.0  0.0 [cpuhp/0]
     21 root      0.0  0.0 [cpuhp/1]
     22 root      0.0  0.0 [idle_inject/1]
     23 root      0.0  0.0 [migration/1]
     24 root      0.2  0.0 [ksoftirqd/1]
     26 root      0.0  0.0 [kworker/1:0H-events_highpri]
     27 root      0.0  0.0 [kworker/u5:0-flush-8:0]
     28 root      0.1  0.0 [kworker/u6:0-events_unbound]
     29 root      0.0  0.0 [kdevtmpfs]
     30 root      0.0  0.0 [kworker/R-inet_]
     31 root      0.0  0.0 [kworker/u5:1-events_unbound]
     32 root      0.0  0.0 [kauditd]
     33 root      0.0  0.0 [khungtaskd]
     34 root      0.0  0.0 [oom_reaper]
     35 root      0.0  0.0 [kworker/u5:2-flush-8:0]
     36 root      0.0  0.0 [kworker/R-write]
     37 root      0.0  0.0 [kcompactd0]
     38 root      0.0  0.0 [ksmd]
     40 root      0.0  0.0 [khugepaged]
     41 root      0.0  0.0 [kworker/R-kinte]
     42 root      0.0  0.0 [kworker/R-kbloc]
     43 root      0.0  0.0 [kworker/R-blkcg]
     44 root      0.0  0.0 [irq/9-acpi]
     45 root      0.0  0.0 [kworker/R-tpm_d]
     46 root      0.0  0.0 [kworker/R-ata_s]
     47 root      0.0  0.0 [kworker/R-md]
     48 root      0.0  0.0 [kworker/R-md_bi]
     49 root      0.0  0.0 [kworker/R-edac-]
     50 root      0.0  0.0 [kworker/R-devfr]
     51 root      0.0  0.0 [watchdogd]
     55 root      0.0  0.0 [kswapd0]
     56 root      0.0  0.0 [ecryptfs-kthread]
     57 root      0.0  0.0 [kworker/R-kthro]
     58 root      0.0  0.0 [kworker/R-acpi_]
     59 root      0.0  0.0 [scsi_eh_0]
     60 root      0.0  0.0 [kworker/R-scsi_]
     61 root      0.0  0.0 [scsi_eh_1]
     62 root      0.0  0.0 [kworker/R-scsi_]
     65 root      0.0  0.0 [kworker/R-mld]
     66 root      0.0  0.0 [kworker/R-ipv6_]
     73 root      0.0  0.0 [kworker/R-kstrp]
     75 root      0.0  0.0 [kworker/u7:0]
     76 root      0.0  0.0 [kworker/u8:0]
     77 root      0.0  0.0 [kworker/u9:0]
     82 root      0.0  0.0 [kworker/R-crypt]
     91 root      0.0  0.0 [kworker/R-charg]
    130 root      0.0  0.0 [kworker/u4:1]
    133 root      0.0  0.0 [kworker/R-mpt_p]
    134 root      0.0  0.0 [kworker/R-mpt/0]
    154 root      0.0  0.0 [scsi_eh_2]
    155 root      0.0  0.0 [kworker/R-scsi_]
    156 root      0.0  0.0 [kworker/u6:2-events_unbound]
    158 root      1.4  0.0 [kworker/1:2-events]
    191 root      0.0  0.0 [kworker/R-raid5]
    228 root      0.0  0.0 [kworker/1:2H-kblockd]
    229 root      0.0  0.0 [jbd2/sda1-8]
    230 root      0.0  0.0 [kworker/R-ext4-]
    297 root      0.0  0.0 [psimon]
    302 root      0.1  1.1 /usr/lib/systemd/systemd-journald
    339 root      0.0  0.0 [kworker/R-kmpat]
    340 root      0.0  0.0 [kworker/R-kmpat]
    362 root      0.2  2.7 /sbin/multipathd -d -s
    365 root      0.1  0.7 /usr/lib/systemd/systemd-udevd
    369 root      1.7  0.0 [kworker/0:4-events]
    371 root      0.0  0.0 [psimon]
    421 root      0.0  0.0 [kworker/0:2H-kblockd]
    439 root      0.0  0.0 [irq/18-vmwgfx]
    440 root      0.0  0.0 [kworker/R-ttm]
    484 root      0.0  0.0 [jbd2/sda16-8]
    485 root      0.0  0.0 [kworker/R-ext4-]
    512 systemd+  0.1  0.9 /usr/lib/systemd/systemd-networkd
    525 root      0.3  0.0 [kworker/1:3-events]
    540 systemd+  0.0  1.3 /usr/lib/systemd/systemd-resolved
    543 systemd+  0.0  0.7 /usr/lib/systemd/systemd-timesyncd
    645 message+  0.0  0.5 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
    649 polkitd   0.0  0.7 /usr/lib/polkit-1/polkitd --no-debug
    655 root      0.0  0.8 /usr/lib/systemd/systemd-logind
```

```
657 root      0.1  1.3 /usr/libexec/udisks2/udisksd
677 syslog    0.0  0.5 /usr/sbin/rsyslogd -n -iNONE
689 root      0.1  2.2 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
719 root      0.0  0.2 /usr/sbin/cron -f -P
722 root      0.0  1.3 /usr/sbin/ModemManager
751 root      0.0  0.1 /sbin/agetty -o -p -- \u --noclear - linux
850 root      0.0  0.8 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
852 root      0.0  0.5 sshd: bella [priv]
859 bella     1.3  0.7 sshd: bella@pts/0
860 bella     0.0  0.5 -bash
901 root      0.4  0.0 [kworker/0:0-cgroup_destroy]
906 root      0.1  0.0 [kworker/u6:3-events_power_efficient]
914 root      0.0  0.0 [kworker/u5:3-flush-8:0]
915 root      2.5  0.0 [kworker/1:0-events]
918 root      0.0  0.0 [kworker/0:1H]
919 root      0.0  0.0 [kworker/1:1H]
920 bella     166  0.4 ps -eo pid,user,pcpu,pmem,cmd
```

Question 5

Building on the ps command used in Question 4, can you add an option to sort the listed processes by their memory usage (`pmem`)?

Command
ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem
Output

```
PID USER     %CPU %MEM CMD
362 root      0.2  2.7 /sbin/multipathd -d -s
689 root      0.1  2.2 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
657 root      0.0  1.3 /usr/libexec/udisks2/udisksd
  1 root      1.1  1.3 /sbin/init
540 systemd+  0.0  1.3 /usr/lib/systemd/systemd-resolved
722 root      0.0  1.3 /usr/sbin/ModemManager
302 root      0.1  1.1 /usr/lib/systemd/systemd-journald
512 systemd+  0.1  0.9 /usr/lib/systemd/systemd-networkd
655 root      0.0  0.8 /usr/lib/systemd/systemd-logind
850 root      0.0  0.8 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
543 systemd+  0.0  0.7 /usr/lib/systemd/systemd-timesyncd
649 polkitd   0.0  0.7 /usr/lib/polkit-1/polkitd --no-debug
365 root      0.1  0.7 /usr/lib/systemd/systemd-udevd
859 bella     1.2  0.7 sshd: bella@pts/0
677 syslog    0.0  0.5 /usr/sbin/rsyslogd -n -iNONE
645 message+  0.0  0.5 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
852 root      0.0  0.5 sshd: bella [priv]
860 bella     0.0  0.5 -bash
924 bella     250  0.4 ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem
719 root      0.0  0.2 /usr/sbin/cron -f -P
751 root      0.0  0.1 /sbin/agetty -o -p -- \u --noclear - linux
  2 root      0.0  0.0 [kthreadd]
  3 root      0.0  0.0 [pool_workqueue_release]
  4 root      0.0  0.0 [kworker/R-rcu_g]
  5 root      0.0  0.0 [kworker/R-rcu_p]
  6 root      0.0  0.0 [kworker/R-slub_]
  7 root      0.0  0.0 [kworker/R-netns]
  9 root      0.3  0.0 [kworker/0:1-events]
 10 root      0.0  0.0 [kworker/0:0H-kblockd]
 11 root      0.0  0.0 [kworker/u4:0-ext4-rsv-conversion]
 12 root      0.0  0.0 [kworker/R-mm_pe]
 13 root      0.0  0.0 [rcu_tasks_kthread]
 14 root      0.0  0.0 [rcu_tasks_rude_kthread]
 15 root      0.0  0.0 [rcu_tasks_trace_kthread]
 16 root      0.0  0.0 [ksoftirqd/0]
 17 root      0.0  0.0 [rcu_preempt]
 18 root      0.0  0.0 [migration/0]
 19 root      0.0  0.0 [idle_inject/0]
 20 root      0.0  0.0 [cpuhp/0]
 21 root      0.0  0.0 [cpuhp/1]
 22 root      0.0  0.0 [idle_inject/1]
```

```
 22 root      0.0  0.0 [idle_inject/1]
 23 root      0.0  0.0 [migration/1]
 24 root      0.2  0.0 [ksoftirqd/1]
 26 root      0.0  0.0 [kworker/1:0H-events_highpri]
 27 root      0.0  0.0 [kworker/u5:0-flush-8:0]
 28 root      0.0  0.0 [kworker/u6:0-events_unbound]
 29 root      0.0  0.0 [kdevtmpfs]
 30 root      0.0  0.0 [kworker/R-inet_]
 31 root      0.0  0.0 [kworker/u5:1-flush-8:0]
 32 root      0.0  0.0 [kauditd]
 33 root      0.0  0.0 [khungtaskd]
 34 root      0.0  0.0 [oom_reaper]
 35 root      0.0  0.0 [kworker/u5:2-flush-8:0]
 36 root      0.0  0.0 [kworker/R-write]
 37 root      0.0  0.0 [kcompactd0]
 38 root      0.0  0.0 [ksmd]
 40 root      0.0  0.0 [khugepaged]
 41 root      0.0  0.0 [kworker/R-kinte]
 42 root      0.0  0.0 [kworker/R-kbloc]
 43 root      0.0  0.0 [kworker/R-blkcg]
 44 root      0.0  0.0 [irq/9-acpi]
 45 root      0.0  0.0 [kworker/R-tpm_d]
 46 root      0.0  0.0 [kworker/R-ata_s]
 47 root      0.0  0.0 [kworker/R-md]
 48 root      0.0  0.0 [kworker/R-md_bi]
 49 root      0.0  0.0 [kworker/R-edac-]
 50 root      0.0  0.0 [kworker/R-devfr]
 51 root      0.0  0.0 [watchdogd]
 55 root      0.0  0.0 [kswapd0]
 56 root      0.0  0.0 [ecryptfs-kthread]
 57 root      0.0  0.0 [kworker/R-kthro]
 58 root      0.0  0.0 [kworker/R-acpi_]
 59 root      0.0  0.0 [scsi_eh_0]
 60 root      0.0  0.0 [kworker/R-scsi_]
 61 root      0.0  0.0 [scsi_eh_1]
 62 root      0.0  0.0 [kworker/R-scsi_]
 65 root      0.0  0.0 [kworker/R-mld]
 66 root      0.0  0.0 [kworker/R-ipv6_]
 73 root      0.0  0.0 [kworker/R-kstrp]
 75 root      0.0  0.0 [kworker/u7:0]
 76 root      0.0  0.0 [kworker/u8:0]
 77 root      0.0  0.0 [kworker/u9:0]
 82 root      0.0  0.0 [kworker/R-crypt]
 91 root      0.0  0.0 [kworker/R-charg]
130 root      0.0  0.0 [kworker/u4:1]
133 root      0.0  0.0 [kworker/R-mpt_p]
134 root      0.0  0.0 [kworker/R-mpt/0]
154 root      0.0  0.0 [scsi_eh_2]
155 root      0.0  0.0 [kworker/R-scsi_]
```

```
156 root      0.0  0.0 [kworker/u6:2-events_unbound]
158 root      1.3  0.0 [kworker/1:2-events]
191 root      0.0  0.0 [kworker/R-raid5]
228 root      0.0  0.0 [kworker/1:2H-kblockd]
229 root      0.0  0.0 [jbd2/sda1-8]
230 root      0.0  0.0 [kworker/R-ext4-]
297 root      0.0  0.0 [psimon]
339 root      0.0  0.0 [kworker/R-kmpat]
340 root      0.0  0.0 [kworker/R-kmpat]
369 root      1.7  0.0 [kworker/0:4-events]
371 root      0.0  0.0 [psimon]
421 root      0.0  0.0 [kworker/0:2H-kblockd]
439 root      0.0  0.0 [irq/18-vmwgfx]
440 root      0.0  0.0 [kworker/R-ttm]
484 root      0.0  0.0 [jbd2/sda16-8]
485 root      0.0  0.0 [kworker/R-ext4-]
525 root      0.4  0.0 [kworker/1:3-events]
901 root      0.5  0.0 [kworker/0:0-events]
906 root      0.1  0.0 [kworker/u6:3-events_unbound]
914 root      0.0  0.0 [kworker/u5:3-events_unbound]
915 root      1.2  0.0 [kworker/1:0-events]
918 root      0.0  0.0 [kworker/0:1H]
919 root      0.0  0.0 [kworker/1:1H]
```

Question 6

Construct a command using `ps`, suitable options, and any additional tools to visualize the hierarchical structure (tree-like) of the following processes:

- "mainprocess"
- "subprocess1"
- "subprocess2"

Command
ps -ef --forest | grep -E "mainprocess|subprocess1|subprocess2"
Output
`bella 927 860 50 14:53 pts/0 00:00:00 \_ grep --color=auto -E mainprocess|subprocess1|subprocess2`

Question 7

Use `pstree` command with option that show the number of threads to each process.

Command
pstree -p -c
Output

```
systemd(1)-+-ModemManager(722)-+-{ModemManager}(732)
           |                   |-{ModemManager}(734)
           |                   `-{ModemManager}(740)
           |-agetty(751)
           |-cron(719)
           |-dbus-daemon(645)
           |-multipathd(362)-+-{multipathd}(379)
           |                 |-{multipathd}(381)
           |                 |-{multipathd}(382)
           |                 |-{multipathd}(383)
           |                 |-{multipathd}(384)
           |                 `-{multipathd}(385)
           |-polkitd(649)-+-{polkitd}(693)
           |              |-{polkitd}(696)
           |              `-{polkitd}(697)
           |-rsyslogd(677)-+-{rsyslogd}(712)
           |               |-{rsyslogd}(713)
           |               `-{rsyslogd}(714)
           |-sshd(850)---sshd(852)---sshd(859)---bash(860)---pstree(928)
           |-systemd-journal(302)
           |-systemd-logind(655)
           |-systemd-network(512)
           |-systemd-resolve(540)
           |-systemd-timesyn(543)---{systemd-timesyn}(586)
           |-systemd-udevd(365)
           |-udisksd(657)-+-{udisksd}(692)
           |              |-{udisksd}(694)
           |              |-{udisksd}(702)
           |              |-{udisksd}(724)
           |              `-{udisksd}(728)
           `-unattended-upgr(689)---{unattended-upgr}(788)
```

Question 8

Use `renice` command to change priority level of one of process “subprocess1”.

Command
ps -ef | grep subprocess1
Output
`bella 931 860 50 14:54 pts/0 00:00:00 grep --color=auto subprocess1`

Question 9

Terminate all running processes with the name “mainprocess”.

Command
killall mainprocess
Output
`mainprocess: no process found`

Question 10

Write a short C or Python code (choose only one language) demonstrating multiprocessing with `fork()` and `wait()`. Compile and/or run the code. Show the output.

Source Code:

```
  GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("Fork failed");
        return 1;
    } else if (pid == 0) {
        // Child process
        printf("Child process %d is running\n", getpid());
        sleep(2);
        printf("Child process %d is exiting\n", getpid());
    } else {
        // Parent process
        printf("Parent process %d is waiting for child process %d\n", getpid(), pid);
        wait(NULL);
        printf("Parent process %d resumes execution\n", getpid());
    }

    return 0;
}
```

Output:

```
Parent process 2451 is waiting for child process 2452
Child process 2452 is running
Child process 2452 is exiting
Parent process 2451 resumes execution
```